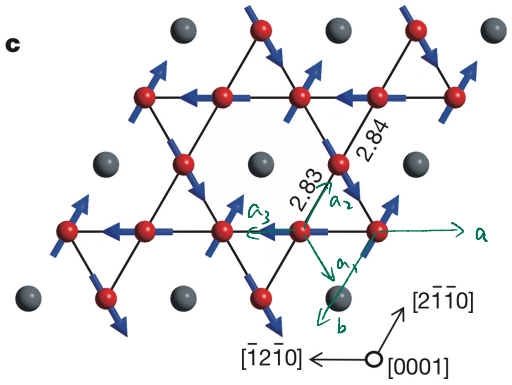
crystal Structure

没有考虑 easy plane anisotropy ?

导致 simulation 电流过大 ?

does the theory of reported anisotropy wrong ? How to introduce easy-plane anisotropy ?

without HM



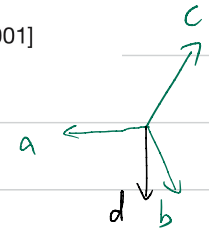
[uvw] ⇒ [U V T W]

$U = \frac{2u-v}{3}$ $T = -(U+V)$

$V = \frac{2v-u}{3}$ $W = w$

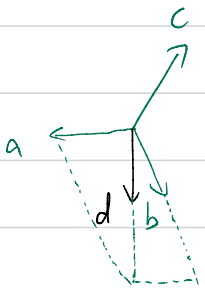
↗ $(\frac{1}{2}, -1, 0)$ ⇒ $\frac{2}{3}, -\frac{5}{6}, -\frac{1}{6}, 0$

全部用 $\vec{a}$, $\vec{b}$ 来表示



c ↗ $[-\frac{1}{2}, -\frac{1}{2}, 0]$

⇒ $[-\frac{1}{3}, -\frac{1}{3}, \frac{2}{3}, 0]$ ⇒ $[2, \bar{1}, \bar{1}, 0]$



a ← $[1, 0, 0]$

⇒ $[2, -1, -1, 0]$ ⇒ $[\bar{1}, 2, \bar{1}, 0]$

d ↓ $[1, 2, 0]$

⇒ $[0, 1, -1, 0]$ ⇒ $[\bar{1}, 0, 1, 0]$

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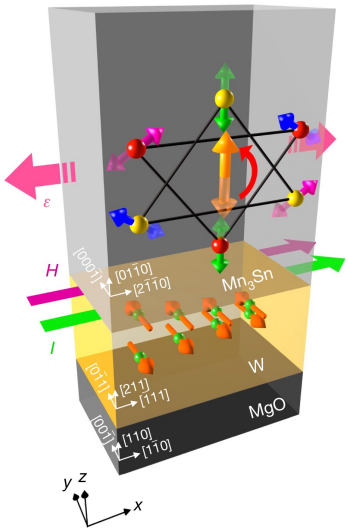
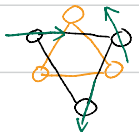
with HM [4]



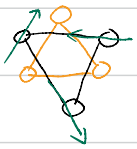
from the substrate side. The X-ray spectra reveal that the Mn_3Sn layer has a hexagonal DO_{19} structure, and it is selectively oriented as MgO(110)[001] || Mn_3Sn(01$\bar{1}$0)[0001] (Fig. 1a, Methods and Extended Data Fig. 1). This orientation is also confirmed by cross-sectional transmission electron microscopy (TEM) (Extended Data Fig. 2b–d). More-

Stable state

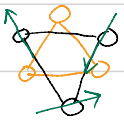
[4] ①



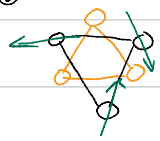
[4] ②



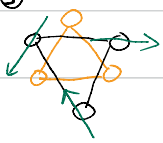
[4] ④



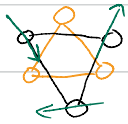
[4] ⑥



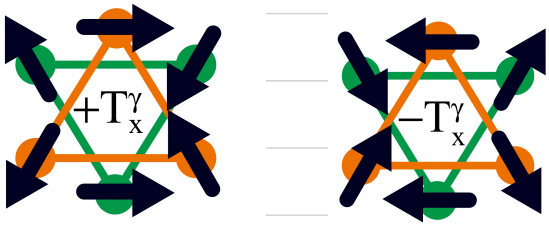
[4] ⑤



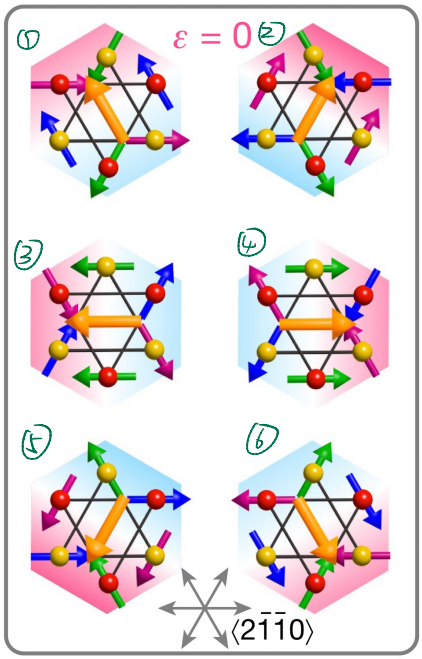
[4] ③



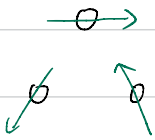
[1]



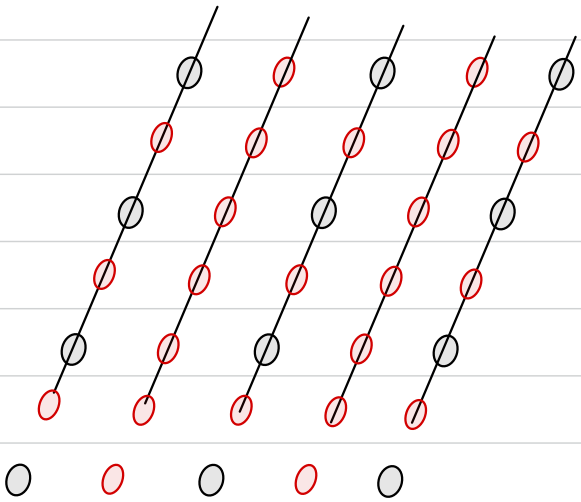
[4]

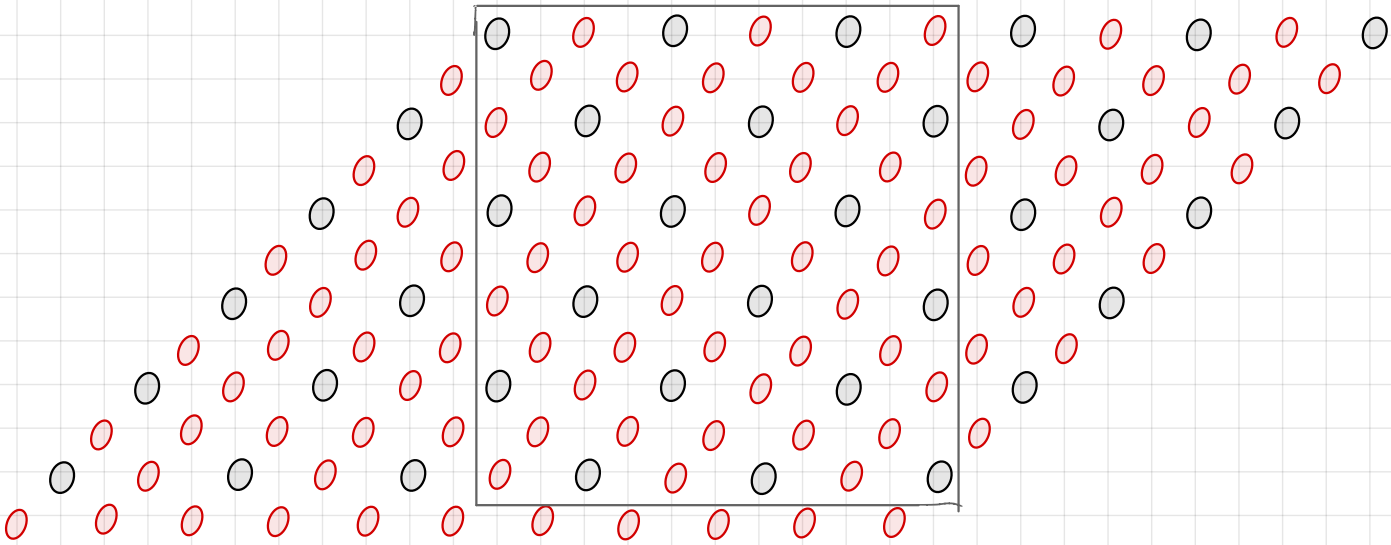


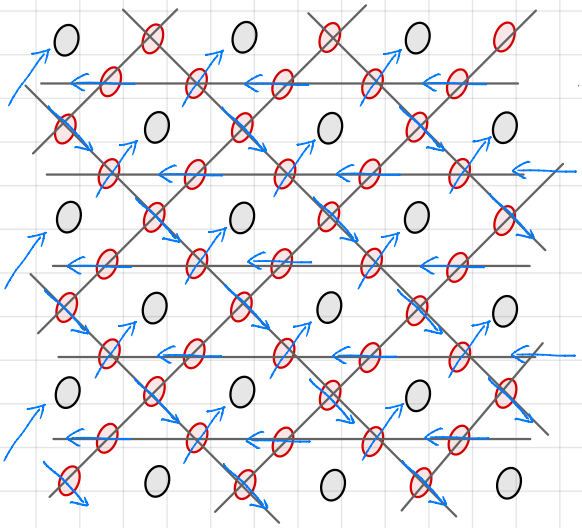
[2] Fig S1 (d)



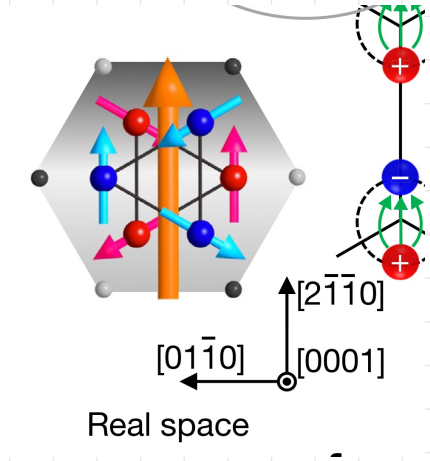
[3]

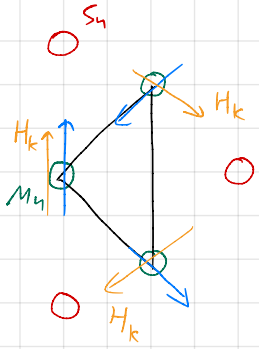




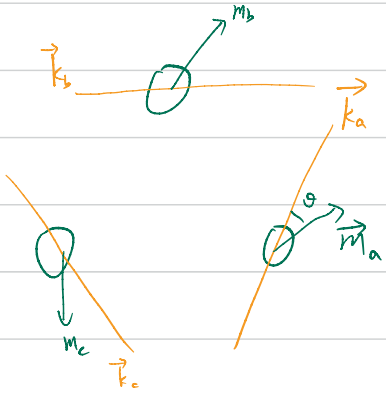


[3]





Anisotropy



$$E = k_u \sin^2\theta = k_u(1-\cos^2\theta) = k_u - k_u(\vec{k}_a \cdot \vec{m})^2$$

$$\Rightarrow \vec{H}_a = -\frac{\partial E}{\partial \vec{m}} = \frac{k_u(\vec{k}_a \cdot \vec{m})^2}{\partial \vec{m}}$$

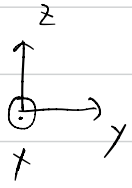
$$= 2k_u(\vec{k}_a \cdot \vec{m}_a)\cdot \vec{k}_a$$

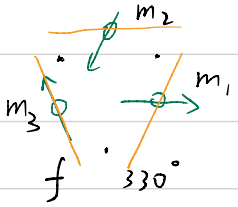
similarly $\vec{H}_b = 2k_u(\vec{k}_b \cdot \vec{m}_b)\cdot \vec{k}_b$

$\vec{H}_c = 2k_u(\vec{k}_c \cdot \vec{m}_c)\cdot \vec{k}_c$

zhengde paper

Configuration II





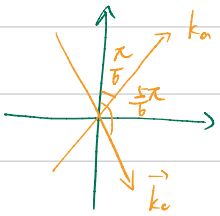
P ⊙ → I//H

$[\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta]$

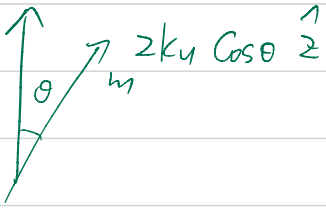
$\vec{k}_1 = \vec{k}_a = [0, \sin\frac{\pi}{6}, \cos\frac{\pi}{6}]$ with $\theta = \frac{\pi}{6}, \varphi = \frac{\pi}{2}$

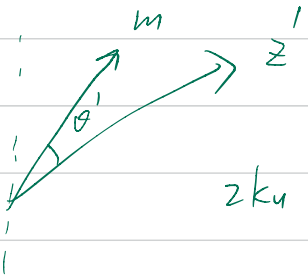
$\vec{k}_2 = \vec{k}_b = [0, \sin\frac{3\pi}{2}, \cos\frac{3\pi}{2}]$ with $\theta = \frac{3\pi}{2}, \varphi = \frac{\pi}{2}$

$\vec{k}_3 = \vec{k}_c = [0, \sin\frac{5\pi}{6}, \cos\frac{5\pi}{6}]$ with $\theta = \frac{5}{6}\pi, \varphi = \frac{\pi}{2}$



anisotropy - old discussion





$2k_u \cos\theta' \, \hat{z}'$

$2k_u \cos(45-\theta) \frac{\sqrt{2}}{2} (\hat{z} + \hat{x})$

$= \sqrt{2}k_u (\cos 45^\circ \cos\theta + \sin 45^\circ \sin\theta) \frac{\sqrt{2}}{2} (\hat{z} + \hat{x})$

$= (k_u \cos\theta + k_u \sin\theta)(\hat{z} + \hat{x})$

$=$

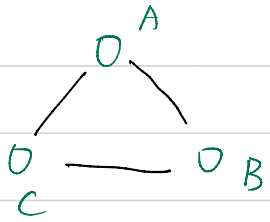
② ③

$$\vec{H}_a = \frac{\partial \left[-k (\vec{k}_a \cdot \vec{m}_a)^2 \right]}{M_s \, \partial \vec{m}_a} = -\frac{2k}{M_s} (\vec{k}_a \cdot \vec{m}_a) \cdot \vec{k}_a \quad \checkmark$$

$$1 - \cos^2\theta = \left[1 - (\hat{z} \cdot m)^2 \right] \qquad 2k \cos(\theta) \, \hat{z} \quad \checkmark$$

$$H_a = \frac{2k_u}{M_s} \cos\theta \, \hat{z}$$

DMI



[5] $U = D\,\hat{e}_z \sum_{\langle \mu\nu \rangle} \vec{m}_\mu \times \vec{m}_\nu$

$\langle \mu\nu \rangle$ indicates sum over $(\mu\nu) = (A,B), (B,C)$ and (C,A)

$$U = D\hat{e}_z \cdot \left(\vec{m}_A \times \vec{m}_B + \vec{m}_B \times \vec{m}_C + \vec{m}_C \times \vec{m}_A \right)$$

$\Rightarrow$

$$U = J_0 \sum_{\langle \mu\nu \rangle} \vec{m}_\mu \cdot \vec{m}_\nu + D\hat{e}_z \sum \vec{m}_\mu \times \vec{m}_\nu - k \sum_{\mu = A,B,C} \left(\vec{m}_\mu \cdot \hat{e}_{k,\mu} \right)^2$$

$$= J_0 \left(\vec{m}_A \cdot \vec{m}_B + \vec{m}_B \cdot \vec{m}_C + \vec{m}_C \cdot \vec{m}_A \right)$$
$$+ D\hat{e}_z \left(\vec{m}_A \times \vec{m}_B + \vec{m}_B \times \vec{m}_C + \vec{m}_C \times \vec{m}_A \right)$$

$$E = D\hat{x}\cdot(\vec{m}_1\times\vec{m}_2 + \vec{m}_2\times\vec{m}_3 + \vec{m}_3\times\vec{m}_1)$$

Assume D along $\hat{x}$ direction

$$\equiv D[\vec{m}_1\cdot(\vec{m}_2\times\hat{x}) + \hat{x}(\vec{m}_2\times\vec{m}_3) + \vec{m}_1\cdot(\hat{x}\times\vec{m}_3)]$$

$$a(b\times c) = b\cdot(c\times a) = c\cdot(a\times b)$$

$$\equiv D[\vec{m}_2\cdot(\hat{x}\times\vec{m}_1) + \vec{m}_2(\vec{m}_3\times\hat{x}) + \hat{x}(\vec{m}_3\times\vec{m}_1)]$$

$$\equiv D[\hat{x}\cdot(\vec{m}_1\times\vec{m}_2) + \vec{m}_3\cdot(\hat{x}\times\vec{m}_2) + \vec{m}_3(\vec{m}_1\times\hat{x})]$$

$$H_{DMI,1} = -\frac{\partial E}{\partial m_1}$$

$$= -D(\vec{m}_2\times\hat{x} + \hat{x}\times\vec{m}_3)$$

$$= (m_{2x}, m_{2y}, m_{2z})\times\hat{x} + \hat{x}\times(m_{3x}, m_{3y}, m_{3z})$$

$$\begin{vmatrix} x & y & z \\ m_{2x} & m_{2y} & m_{2z} \\ 1 & 0 & 0 \end{vmatrix} = \hat{y}\,m_{2z} - m_{2y}\hat{z}$$

$$= -D(\hat{y}\,m_{2z} - m_{2y}\hat{z} - \hat{y}\,m_{3z} + \hat{z}\,m_{3y})$$

$$= -D[0,\ m_{2z} - m_{3z},\ -m_{2y} + m_{3y}]$$

$$\begin{vmatrix} x & y & z \\ 1 & 0 & 0 \\ m_{3x} & m_{3y} & m_{3z} \end{vmatrix} = -\hat{y}\,m_{3z} + \hat{z}\,m_{3y}$$

$$H_{DMI,2} = -\frac{\partial E}{\partial m_2}$$

$$= -D(\hat{x}\times\vec{m}_1 + \vec{m}_3\times\hat{x})$$

$$= -D(\hat{y}\,m_{3z} - m_{3y}\hat{z} - \hat{y}\,m_{1z} + \hat{z}\,m_{1y})$$

$$= -D[0,\ m_{3z} - m_{1z},\ m_{1y} - m_{3y}]$$

$$H_{DMI,3} = -\frac{\partial E}{\partial m_3}$$

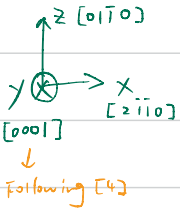
$$= -D(\hat{x}\times\vec{m}_2 + \vec{m}_1\times\hat{x})$$

$$= -D(\hat{y}\,m_{1z} - m_{1y}\hat{z} - \hat{y}\,m_{2z} + \hat{z}\,m_{2y})$$

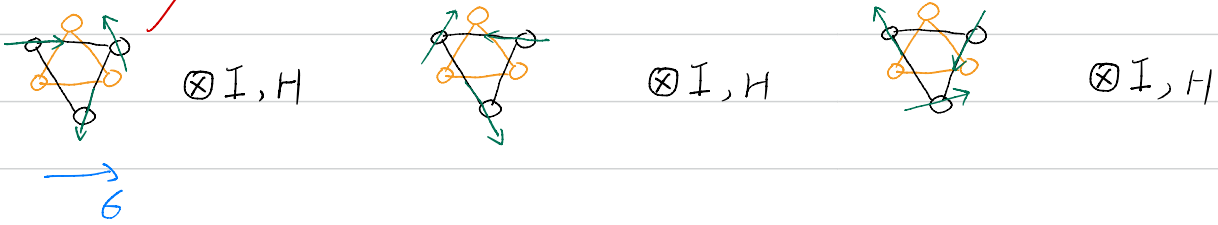
$$= -D[0,\ m_{1z} - m_{2z},\ m_{2y} - m_{1y}]$$

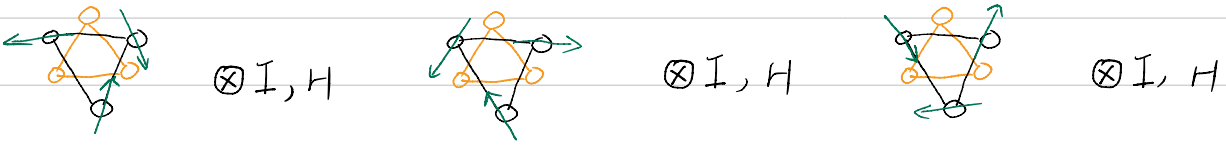
SOT switching

算黑的3个和orange的3个会有区别吗？（how project, rotate plane）

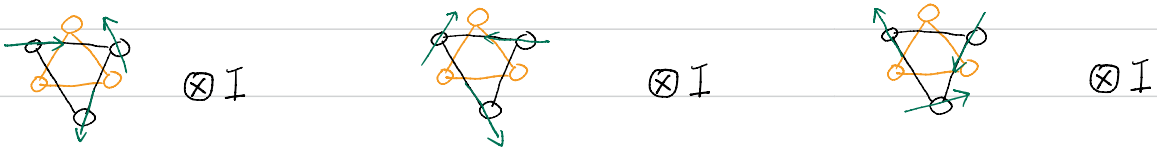


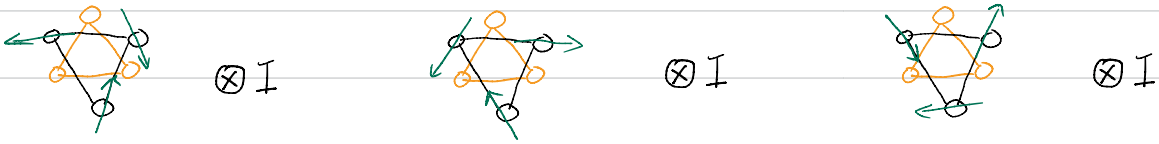
Configuration (a) in [3], I ⊥ kagome plane, G // kagome plane



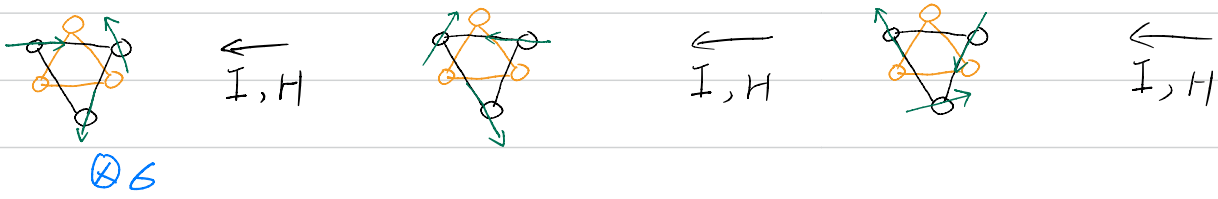


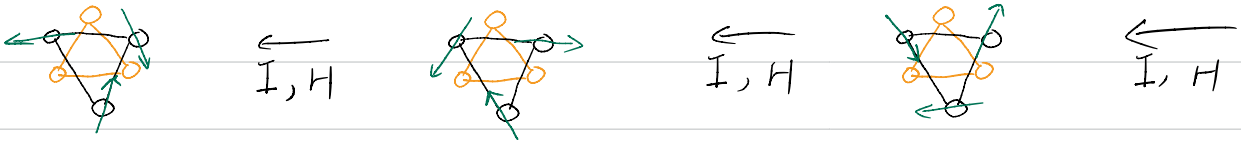
Configuration (a) in [3], I ⊥ kagome plane, G // kagome plane



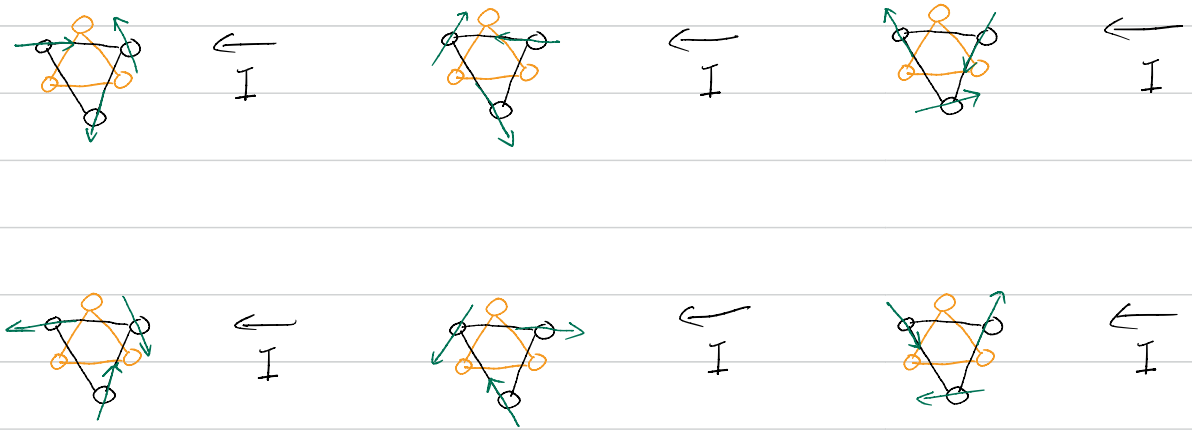


Configuration (b) in [3], I // kagome plane, G ⊥ kagome plane

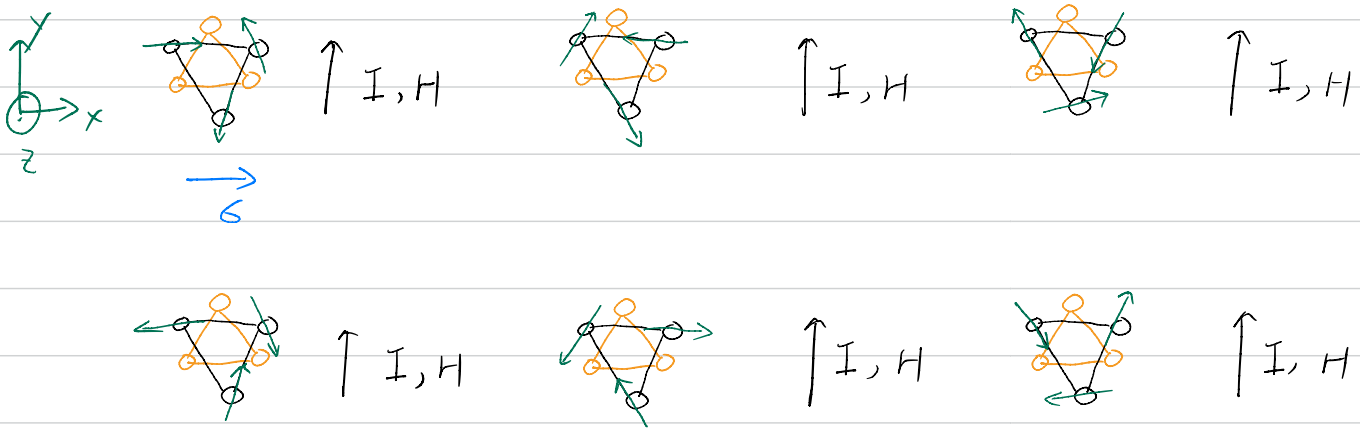




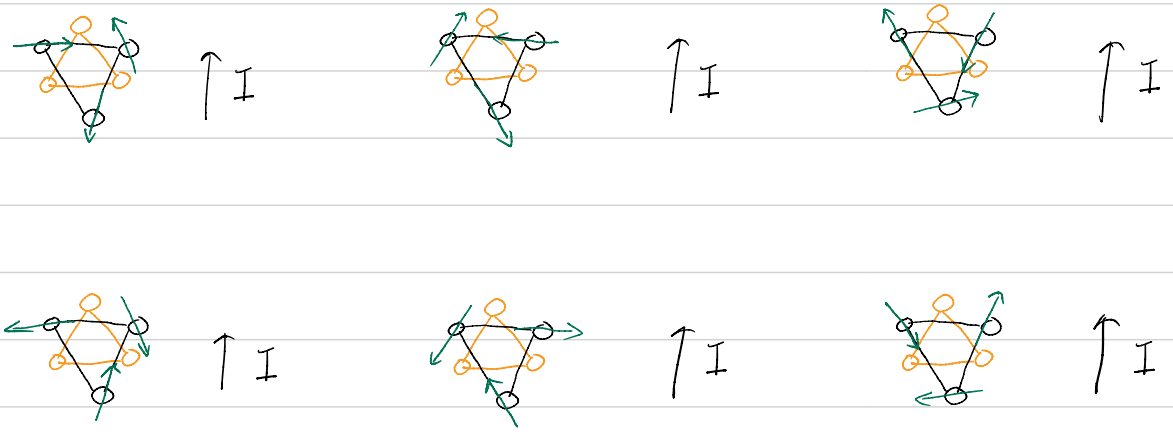
Configuration (b) in [3], I // kagome plane, G ⊥ kagome plane



Configuration (c) in [3], I // kagome plane, G // kagome plane



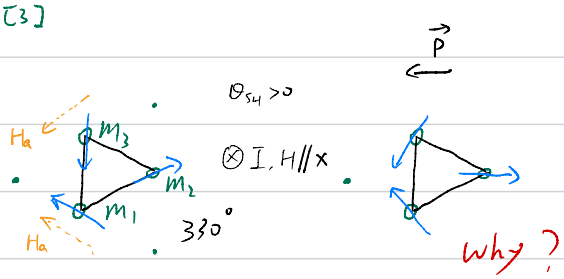
Configuration (c) in [3], I // kagome plane, G // kagome plane



$J_s = \theta_{SH} J_c$

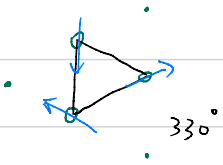
$-z \quad +y \quad +x$

[3]

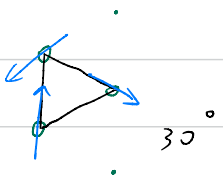


why ?

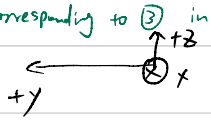
$+J$ $H_x > 0$

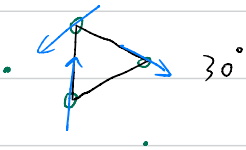


$+J$ $H_x < 0$

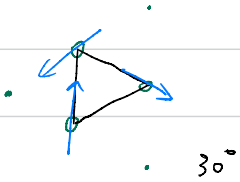


Corresponding to ③ in Page 1

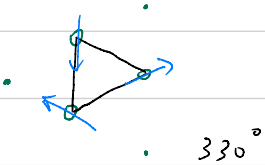




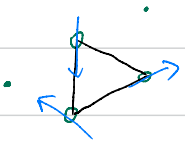
$-J$ $+H_x$

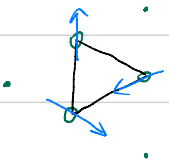


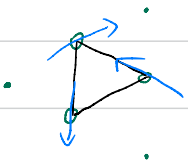
$-J$ $-H_x$

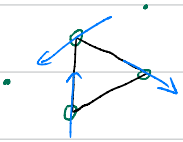


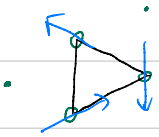
Corresponding to ⑤ in Page 1

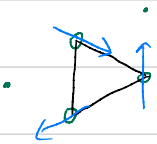


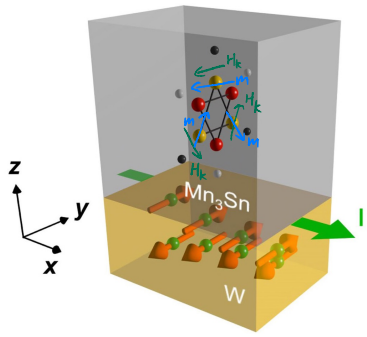




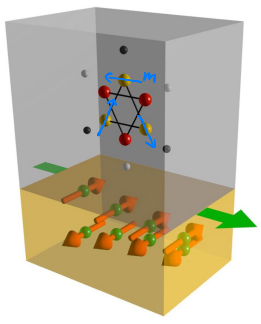








switching the same as STT

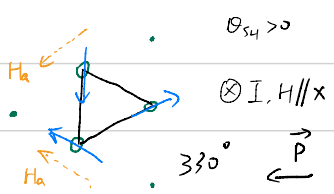


type x switching

To compare with zhengde results of Conf. I, we need to transform the axis

$J_s = \theta\ 6\times J_c$

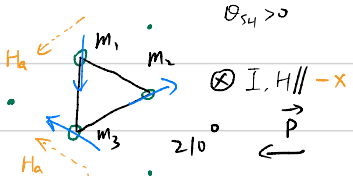
$-z \quad +y \quad +x$



$\Rightarrow$

$J_s = \theta\ 6\times J_c$

$y \quad -z \quad -x$

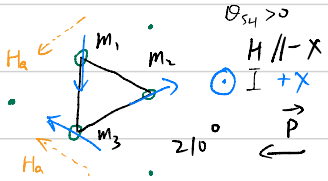


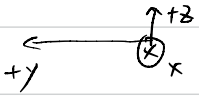
$\Rightarrow$

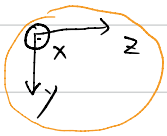
-y 变为 +y，可以看成 J_c 反号

$J_s = \theta\ 6\times J_c$

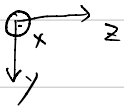
$+y \quad -z \quad +x$

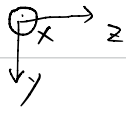




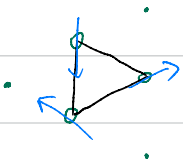


坐标系改变 ⇒ x 反号

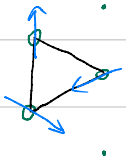




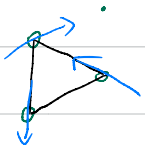
$\Rightarrow$



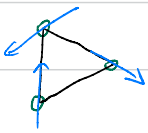
210° d



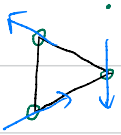
30° a



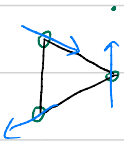
330° f



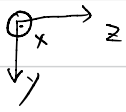
150° c



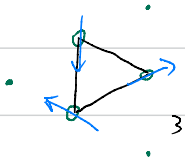
90° b



270° e

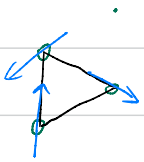


$+J$ $H_x > 0$



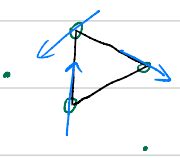
330°

$+J$ $H_x < 0$



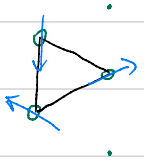
30°

$-J$ $+H_x$



30°

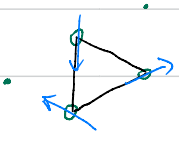
$-J$ $-H_x$



330°

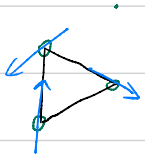
I 不变
H 反号 ⟹

$J_x > 0$ $H_x < 0$



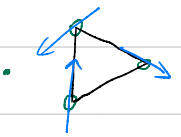
210°

$J_x > 0$ $H_x > 0$



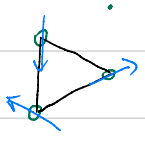
150°

$J_x < 0$ $H_x < 0$



150°

$J_x < 0$ $H_x > 0$

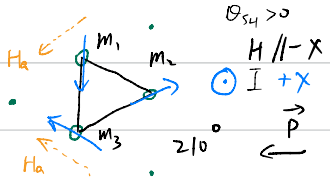


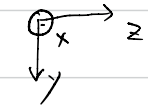
210°

Summary

$J_s = \theta \, G \times J_c$

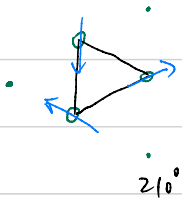
$+y \quad -z \quad +x$



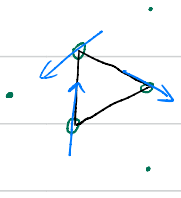


$\Rightarrow$

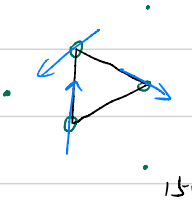
$J_x > 0$
$H_x < 0$



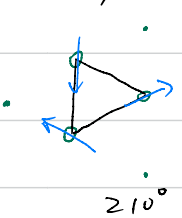
$J_x > 0$
$H_x > 0$



$J_x < 0$
$H_x < 0$



$J_x < 0$
$H_x > 0$

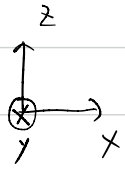


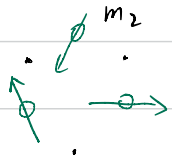
Zhengde paper

Configuration I

$J_s = \theta_G \times J_c$

z x y

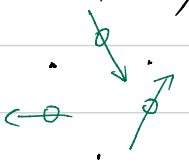




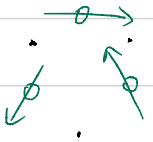
330° f

P

⊗ I//H

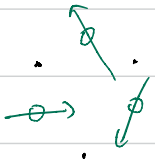


30° a

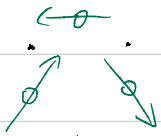


90° b

150° c



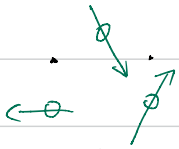
210° d



270° e

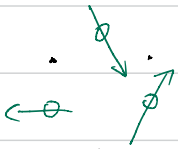
$-J_c$, $+H_{ext}$
$+J_c$, $-H_{ext}$

$-J_c$, $-H_{ext}$
$+J_c$, $+H_{ext}$

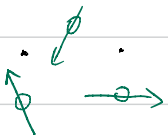


30° a

⇒



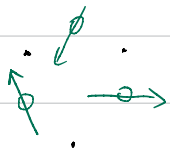
30° a



330° f

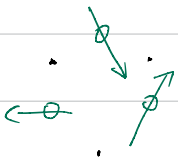
$-J_c$, $+H_{ext}$
$+J_c$, $-H_{ext}$

$-J_c$, $-H_{ext}$
$+J_c$, $+H_{ext}$

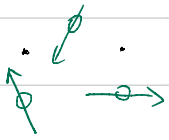


330° f

⇒



30° a



330° f

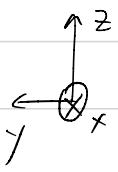
To unify the axis, we need to transform the axis

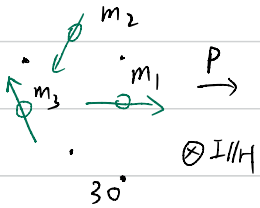
Zhengde paper

$J_s = \theta \; 6\times J_c$

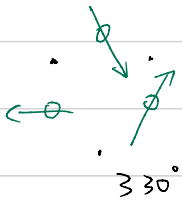
$+z \quad -y \quad +x$

Configuration I

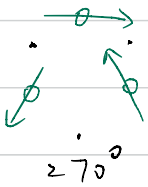




30°

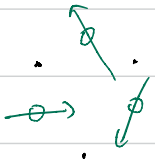


330°

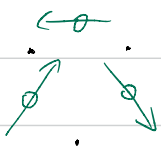


270°

210°

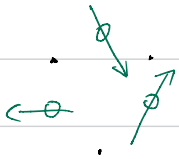


150°



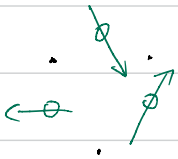
90°

$-J_c, +H_{ext}$
$+J_c, -H_{ext}$



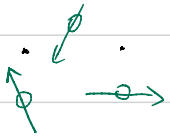
330°

⇒



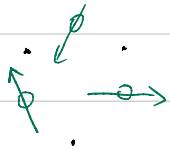
330°

$-J_c, -H_{ext}$
$+J_c, +H_{ext}$



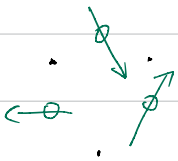
30°

$-J_c, +H_{ext}$
$+J_c, -H_{ext}$



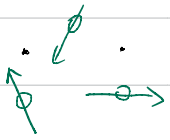
30°

⇒



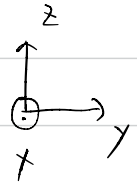
330°

$-J_c, -H_{ext}$
$+J_c, +H_{ext}$

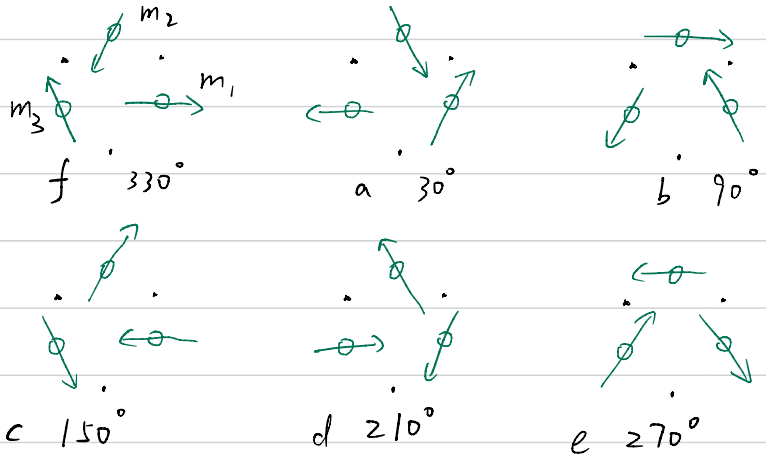


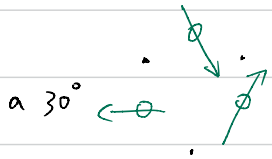
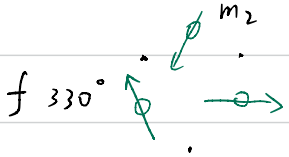
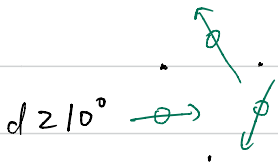
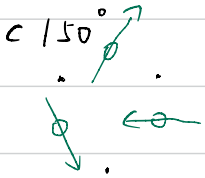
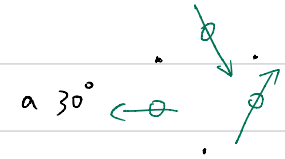
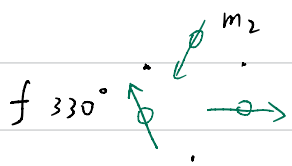
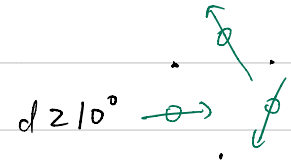
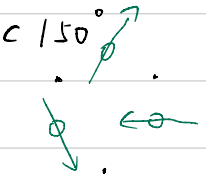
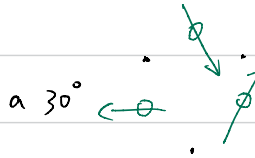
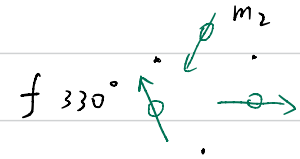
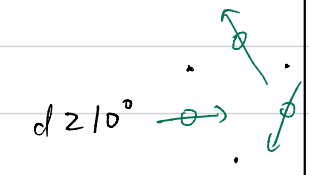
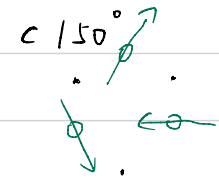
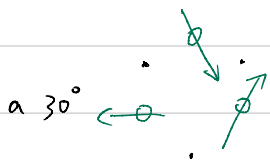
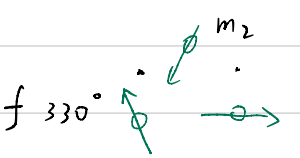
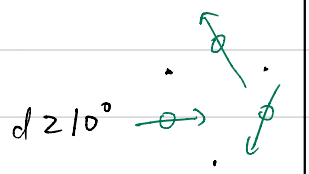
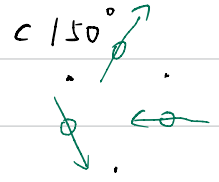
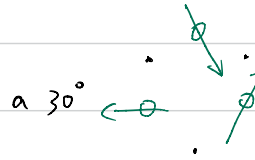
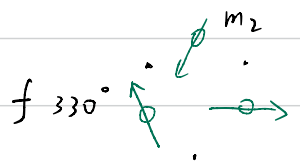
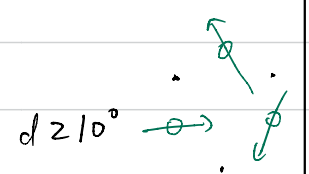
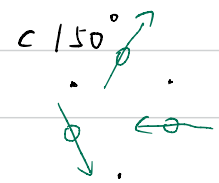
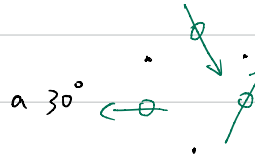
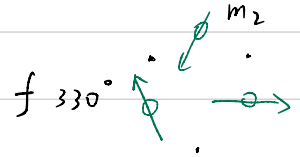
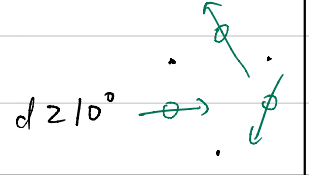
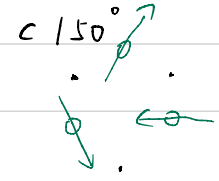
30°

Configuration II



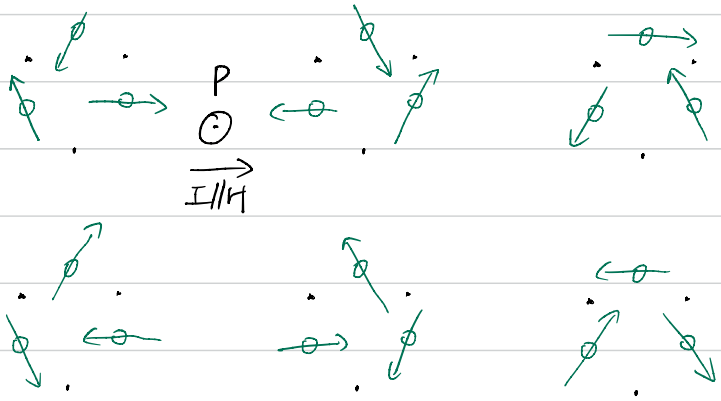
P ⊙ → I//H



	J_{+y}, H_{+y}	J_{-y}, H_{-y}	$J_{+y}\ H_{-y}$	$J_{-y}\ H_{+y}$
a 30°				
b 90°				
c 150°				
d 210°				
e 270°				
f 330°				

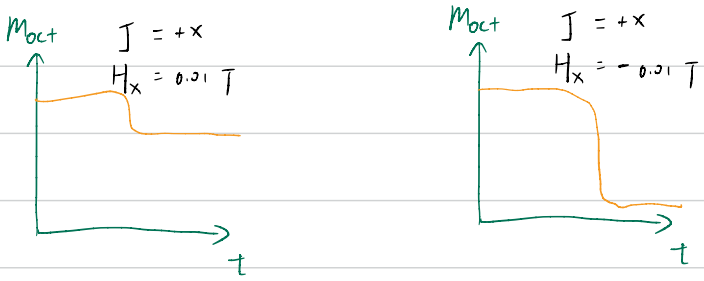
$J_s = \theta_G \times J_c$

$z \quad x \quad y$



H_x small oscillation

H_x mediem



H_x large

it evolves to a new state (m is sensitive to in-plane H_{eff})

Parameters

$\mu_B = 9.274e-24$ J/T T: $\frac{V \cdot s}{m^2}$

$e = 1.602e-19$ C

[3]

$M_S = 3\mu_B$

$T_{ia} = \frac{\hbar V J_0 \theta_{SH}}{2e m_s d}$ $m_s = 6 M_S$ per unit cell Volume

$J = 23$ meV $D = 1.6$ meV $k = 0.17$ meV $\alpha = 0.003$ $\theta_{SH} = 0.1$

$d = 40$ nm $J_0 = 6 \times 10^{14}$ A/m^2 $H_{ext} = 0.1$ T

$H = J \sum \vec{m}_{ia} \vec{m}_{ib}$

$H_{eff} = -\frac{1}{M_S} \frac{\partial H}{\partial m_{ia}}$

$H_{ex} = \frac{23 \times 10^{-3} \times 1.602e-19 \text{ J}}{3 \times 9.274e-24 \text{ J/T}} = 132.43$ T

$\frac{\hbar J_0 \theta_{SH}}{2e m_s d} = \frac{eV \cdot s \cdot A/m^2}{e \; A/m \cdot m} = \frac{V \cdot s}{m^2} = T$

$m_s = 18 \mu_B = $ J/T

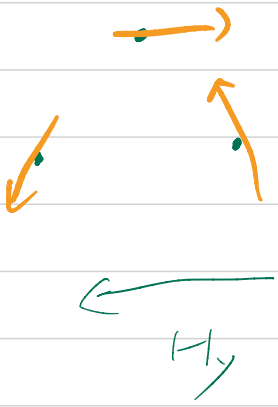
$M_S = \frac{m_s}{V} = \frac{J/T}{m^3} = \frac{VAs \, m^2}{V \cdot s \, m^3 m} = A/m$

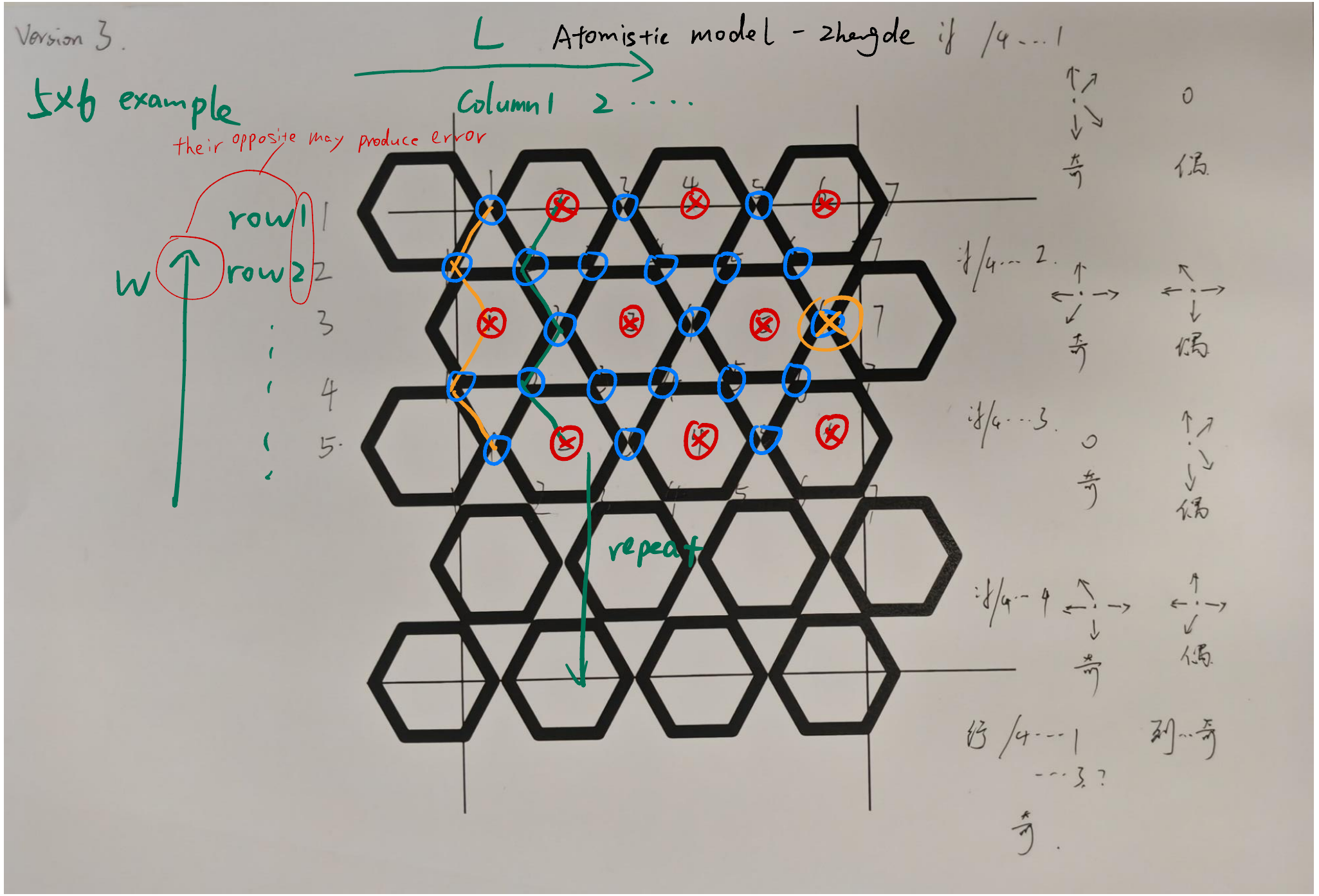
[5] $H_{SOT} = \frac{V \hbar \theta_{SH} J}{2e M_s d}$ $d = 10$ nm

$M_S = 1.26$ T

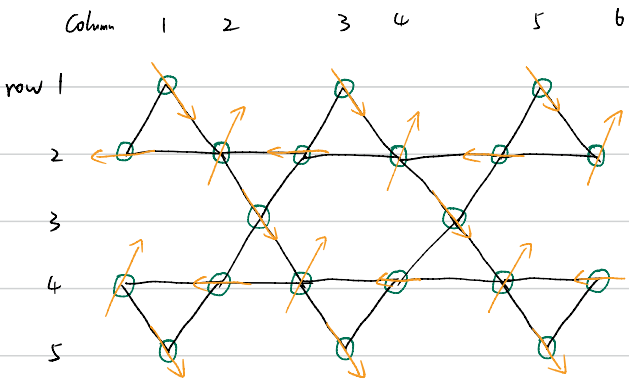
Configuration I

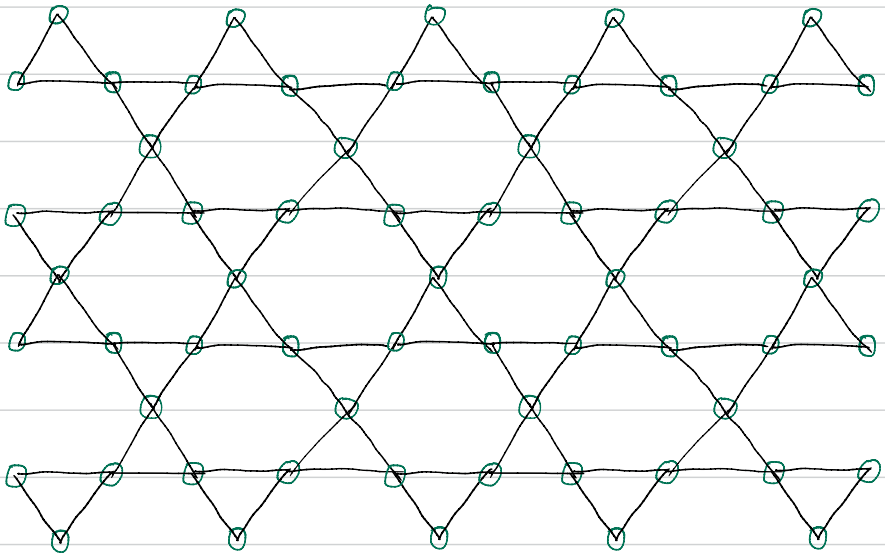
[3]




Version 3.
L Atomistic model - zhangde if /4...1
5x6 example
Column1 2 ....
their opposite may produce error
row1
row2
W
repeat
0
奇
偶
if/4...2.
if/4...3.
if/4...4
行/4...1
...3?
列...奇
奇.

最近邻



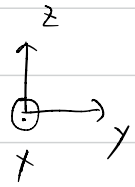


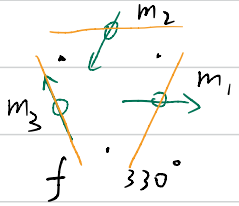
Atom_type

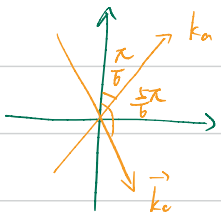
1	3	1	3	1	3
1	1	1	1	1	1
3	1	3	1	3	3
1	1	1	1	1	1
1	3	1	3	1	3

zhengde paper

Configuration II







anisotropy - y, z

$\frac{3}{2}\pi$	X	$\frac{3}{2}\pi$	X	$\frac{3}{2}\pi$	X
$\frac{5}{6}\pi$	$\frac{\pi}{6}$	$\frac{5}{6}\pi$	$\frac{\pi}{6}$	$\frac{5}{6}\pi$	$\frac{\pi}{6}$
X	$\frac{3}{2}\pi$	X	$\frac{3}{2}\pi$	X	X
$\frac{\pi}{6}$	$\frac{5}{6}\pi$	$\frac{\pi}{6}$	$\frac{5}{6}\pi$	$\frac{\pi}{6}$	$\frac{5}{6}\pi$
$\frac{3}{2}\pi$	X	$\frac{3}{2}\pi$	X	$\frac{3}{2}\pi$	X

A next L (最后一列为0)

奇数行全0	X	0	X	0	X
偶数行全1	A	A	A	A	0
	0	X	0	X	X
	A	A	A	A	0
	X	0	X	0	X

A prev L (第1列为0)

奇数行全0	X	0	X	0	X
偶数行全1	A	A	A	A	A
奇数行全0	0	X	0	X	X
偶数行全1	A	A	A	A	A
	X	0	X	0	X

A next W

第一行全0	X	0	X	0	X
第2行奇数列为1	A	A	A	A	0
奇数行全1	X	A	X	A	X
第4行偶数列为1	A	A	A	A	0
	X	A	X	A	X

A prev W

奇数行全A	X	A	X	A	X
第2行偶数列为1	A	A	A	A	A
	X	A	X	A	X
第4行奇数列为1	A	A	A	A	A
最后一行全0	X	X	X	X	X

A next W left (第1列总为0)

第一行总为0	X	0	X	0	X
第2行偶数列为A	A	A	A	A	A
奇数行为0	X	0	X	0	X
第4行奇数列为A	A	A	A	A	A
奇数行为0	X	0	X	0	X

A prev W left (第一列总为0)

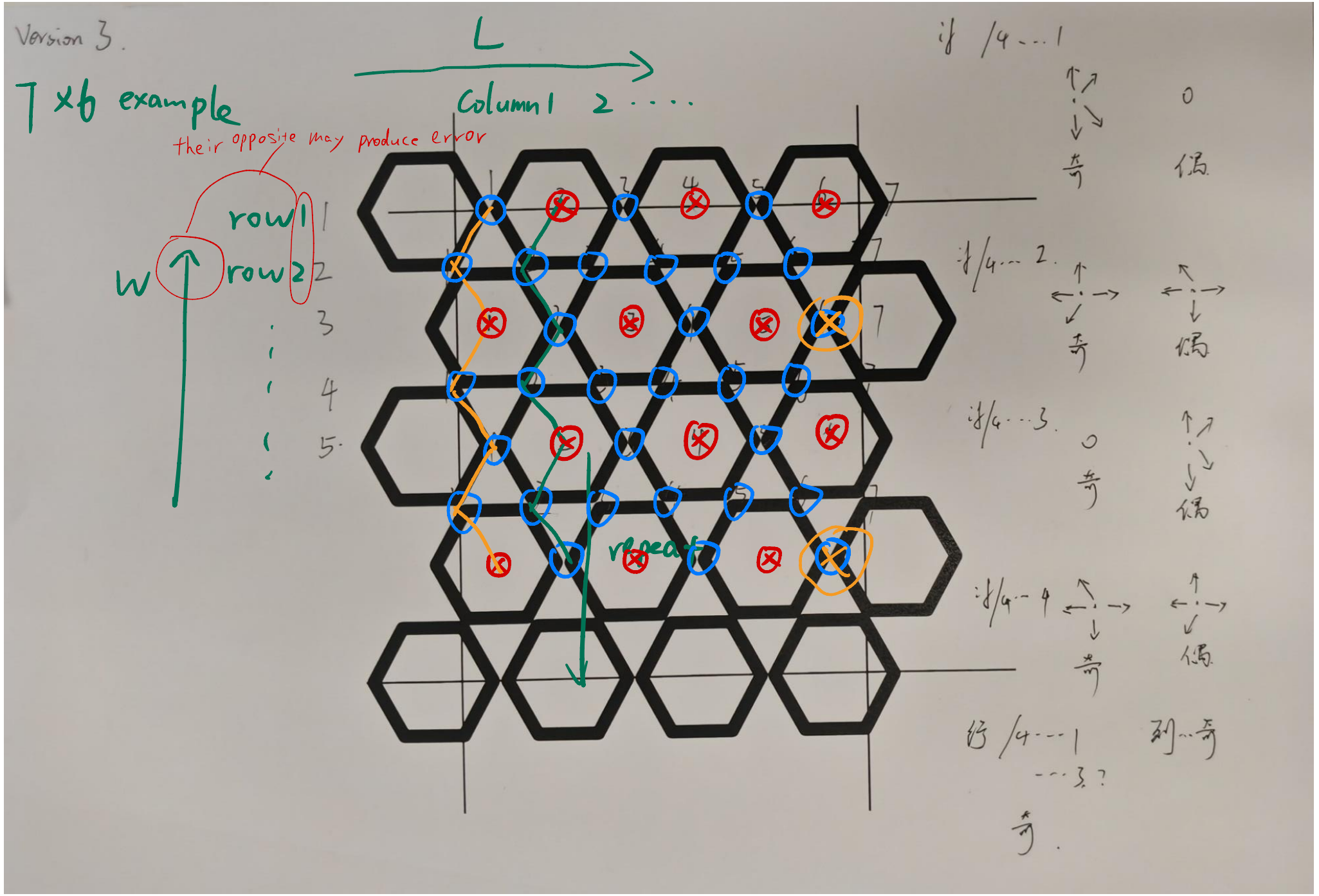
	X	0	X	0	X
第2行奇数列有prev_left	0	A	0	A	0
奇数行总有prev_left	X	0	X	0	X
第4行偶数列有prev_left	A	A	A	A	A
最后一行总为0	X	0	X	0	X

A next W right (最后一列总为0)

第一行总为0	X	0	X	0	X
	0	0	0	0	0
奇数行总为1	X	A	X	A	X
偶数行总为0	0	0	0	0	0
	A	X	A	X	A

A prev W right (最后一列总为0)

奇数行总为A	X	A	X	A	X
偶数行总为0	0	0	0	0	0
奇数行总为A	X	A	X	A	X
第4行总为0	0	0	0	0	0
最后一行总为0	X	0	X	0	X


Version 3.
L
Column 1 2 ....
Txb example
their opposite may produce error
row1
row2
W
repeat
if /4 ... 1
0
奇
偶
if/4 ... 2.
奇
偶
if/4 ... 3.
0
奇
偶
if/4 ... 4
奇
偶
行 /4 ... 1
... 3?
列...奇
奇.

最近邻

Refs

[1] 2017 PRB suzuki

[2] 2015 nature Satoru Nakatsuji

[3] 2020 nature Hanshen Tsai

[4] 2022 nature Tomoya Higo

[5] 2021 nat mat Yutaro Takeuchi